

HACKCANTON SEASON #2 · GRAND FINAL · 5 AUGUST 2026

tirai.

The confidential multi-dealer RFQ / OTC desk, built native on Canton.
Sealed quotes, atomic delivery-versus-payment, post-trade audit.

You whisper quotes. The market hears nothing.

Asking for a price **is** the information

A bank wants to sell fifty million of bonds. Before it trades, it must ask dealers for a price.

**The moment anyone sees you asking,
they know your size and your direction.**

THE TWO THINGS INSTITUTIONS DO TODAY

Privacy, or proof. Never both.

TRADE IN PUBLIC

Everything leaks

On a transparent chain your request is a transaction and every competing quote is a transaction. Rivals simply read them. Front-running and adverse selection are structural, not incidental.

TRADE ON THE PHONE

Nothing is provable

A voice broker or a chat room is private — and leaves no record. Six months later you cannot prove to compliance that you got the best price available. Nothing is atomic.

**Which is why block trading in 2026
still happens on the telephone.**

TIRAI GIVES YOU BOTH

The dealer terminal, on the ledger

01 · RFQ

Request, unseen

A buyer asks a chosen dealer panel. The market never sees the enquiry.

02 · SEALED QUOTE

Whispered

Each dealer answers the buyer alone. Rivals never receive the contract.

03 · ESCROW

Committed

Quoting locks the dealer's bond. A price is a commitment, not a bluff.

04 · AWARD

Second price

Cheapest ask wins and is paid the runner-up's price. Honest quoting dominates.

05 · SETTLE

Atomic DvP

Bond and cash move in one transaction. The regulator gets the report.

Three execution rails — **reverse-Vickrey auction**, **direct bilateral OTC**, **partial fills** on both. Multi-instrument baskets settle as one atomic multi-leg trade.

You sign in as **one identity** and the home screen is the **book**: every request that party is allowed to see, and nothing else.

NOT HIDDEN BY THE INTERFACE

Dealer B's node **never received** dealer A's quote

Dealer A's session — **its own sealed ask**

Active RFQs

every request this party can see — rival dealers see only the ones they were invited to

All 62

Mine 0

For me 62

UST10Y x

1 of 62 shown

REQUEST	INSTRUMENT	QUANTITY	MODE	COUNTERPARTY	CASH LEG	PRICE	SEALED QUOTES	STATUS
002f223c	UST10Y	1,000	Auction	buyer	USDC	1,042,000 best	1	Open Sealed

Dealer B's session — **the number is not there**

Active RFQs

every request this party can see — rival dealers see only the ones they were invited to

All 19

Mine 0

For me 19

UST10Y x

1 of 19 shown

REQUEST	INSTRUMENT	QUANTITY	MODE	COUNTERPARTY	CASH LEG	PRICE	SEALED QUOTES	STATUS
002f223c	UST10Y	1,000	Auction	buyer	USDC	—	0	Awaiting quotes Quote

One request, the same ledger, the same moment — in each dealer's own session. Dealer B is not being shown a redacted view: its participant node never received the quote contract at all.

THE SUPERPOWER

The chain did the hard part.

TO HIDE ONE NUMBER	WHAT IT COSTS YOU
A trusted execution enclave	Hardware trust, attestation, encrypted handles to manage
Zero-knowledge circuits	Circuits to write and audit, a trusted setup, proving time
Threshold encryption	A committee to run, key shares to hold, liveness to guarantee
Homomorphic encryption	Every branch computed blind, and the performance that implies
Canton	None. Sub-transaction privacy is the ledger model.

"Dealer B cannot see dealer A's quote" is a **signatory / observer** declaration.

WHO BUYS IT

Desks whose tickets are big enough to leak

PRIMARY

Trading desks

Fixed-income and crypto-asset desks at banks, asset managers and prop shops moving \$1m–\$100m tickets that cannot afford to signal.

CHANNEL

Hosting venues

Temple, Bron, Console, Canton Loop — they hold the client relationship and the custody; Tirai ships as an embedded app and shares the fee.

MANDATORY THIRD PARTY

The regulator

Not a payer, but the reason a private chat will not do. Post-trade-only visibility is the middle ground supervisors ask for.

Below a million, leakage is a rounding error.

Above it, leakage is the whole cost.

HOW IT MAKES MONEY

A venue fee the settlement can collect

PER-TRADE FEE

Basis points of notional

Built and settling on DevNet. The cut leaves the same atomic transaction as the trade, so if the trade settles the fee is paid — no invoicing, no collection risk. One real settlement: 4,250,000 cleared at 25 bps, 10,625 to the venue, 4,239,375 to the dealer. The *rate* is still unset: blank charges nothing, and the number comes out of a design-partner conversation.

NETWORK REWARDS

CIP-0047 activity markers

The intended second line: featured-app markers on every settlement, so a live desk keeps earning from the volume it clears on top of the fee. *Not built* — there is no marker in the code, and it needs a mainnet app id to mean anything.

Fee on notional means one desk clearing institutional size is a real business — and tokenised bond issuance is the growth curve underneath it.

DIFFERENTIATION

Not a DEX. Not a chat room.

ALTERNATIVE	THE SINGLE AXIS IT FAILS ON
Public order book / on-chain RFQ	Pre-trade confidentiality. The order <i>is</i> the signal.
Voice or chat broker	Provable settlement. Nothing is atomic; best execution is reconstructed from chat logs.
Tradeweb / MarketAxess / bank dark pools	The venue itself sees everything. Confidentiality is an operator's promise.
Other Canton privacy demos	No mechanism. They show Canton can keep a secret; they do not run a market.

Private price discovery — **with a receipt.**

LIVE ON CANTON DEVNET – VERIFIABLE RIGHT NOW

Not a mock-up

2DEVNET PARTICIPANTS
ONE PACKAGE, UPGRADED**50**SETTLED TRADES
+ 5 ATOMIC BASKETS**16**BEST-EXECUTION
ATTESTATIONS**41**DAML TEST
SCRIPTS GREEN

Deployed on HackCanton's own node and the shared 5N validator from the same package, `tirai-desk`, upgraded in place on the validator without stranding a single earlier contract. Hosted read-only desk over live ledger state at tirai.vercel.app.

Honest label: that ledger history was generated by me. It demonstrates the product works — it is not customer traction, and I will not present it as such.

THE TWO PROOF VIEWS

The claim is checked against the live ledger

VERIFY PRIVACY

Each party's real visible set

Not a claim — a query, run against the network on demand.

BEST EXECUTION

Proof without an order book

A buyer, or a dealer defending its pricing, discloses a single sealed quote to the regulator. Sixteen attestations prove the clearing price beat every disclosed rival ask.

Confidential pre-trade. **Provable post-trade.**

✓ Sub-transaction privacy verified on-ledger

Dealer A's node	holds 15 sealed quote(s) — all 15 its own. Rivals' quotes received: 0 .	✓
Dealer B's node	holds 12 sealed quote(s) — all 12 its own. Rivals' quotes received: 0 .	✓
Regulator's node	holds 55 settled trade(s), 0 sealed quotes, 0 live RFQ(s). Pre-trade visibility: none .	✓

YOUR FEEDBACK ASKED FOR REAL EXTERNAL SETTLEMENT

Two assets. Neither of them mine.

The cash leg settles in **real Canton Coin** through the DSO's registry and in **real CBTC** through the DA Utility Registry that BitSafe issues on. Issuers I do not administer and cannot mint into.

CANTON COIN**6 settlements**

Four reverse-Vickrey, two direct OTC. 60,900 CC moved to the dealers that won them, each bond-against-cash in one transaction.

CBTC**0.34 + 0.22**

A faucet grant accepted on-ledger, then spent through the desk: cleared at the Vickrey second price, and hit directly at the ask.

CETH**One field away**

The instrument administrator is the only difference. The rail appears in the desk the day the tokens land, with no release in between.

Nothing in the model is per-asset: `cashInstrument` is any `{admin, id}`, and each registry only differs in where it keeps its off-ledger choice contexts. The desk's Settlement rails view lists every one it can clear on.

THE TEAM

One person. Five days.

WHO

Pugar Huda Mantoro

Final-year undergraduate, building solo. Everything on the previous slides is my own work: the ledger model, the deployer, the desk, the tests, and the two registry integrations.

WHY IT MOVED THIS FAST

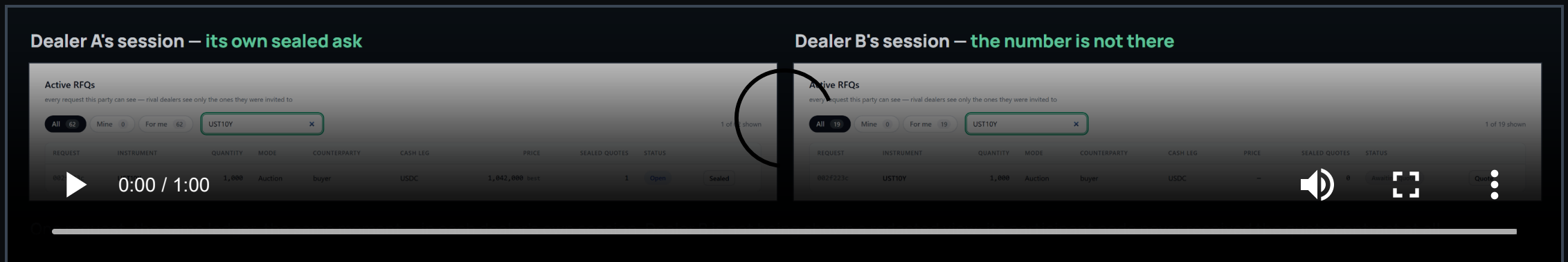
Five days, start to deployed

Repo created 22 July. The CIP-0056 settlement leg landed 23 July. Hosted desk live 24 July. Second participant deployed and privacy-verified 26 July. Since the feedback: real Canton Coin and real CBTC settlement on the ledger.

And a written 90-day validation plan with the stop criteria set *in advance*, because a plan that cannot fail is not a plan.

THE DESK, DRIVEN

Not a slideshow. The product.



Dealer A seals its ask, then the rival's session comes up with no price. Stops itself. Press again to replay.

This page is a still. The film is at tirai.vercel.app/deck (slide 14) and tirai.vercel.app.

A request opened, a dealer's sealed quote, the rival's own session without it, then the privacy verifier and the settlement rails. Screen recording against a live ledger; the only edit is a slow push on the rival's empty price cell.

TIRAI – INDONESIAN FOR “CURTAIN”

Price discovery happens behind it.

I need one design-partner desk. One hour a week for ninety days — not a purchase order.

1 ONE TRADING DESK AS DESIGN PARTNER

2 CETH AND CBTC TEST-TOKEN GRANTS

3 A HOSTING VENUE FOR A SUPERVISED PILOT